Similarly, Pearl found that when the number of hens in a pen increased, the birds struggled to find enough food. As the space between them reduced, the hens laid fewer eggs and, as their fertility rate declined, the rate of population growth slowly stabilized.



Fruit flies are small, common flies that are attracted to ripe fruit and vegetables. They are popular for laboratory studies because they reproduce so quickly and are easy to cultivate.

“Biologists are at the present time in no way likely to suffer ostracism if they venture to study human problems.”

Raymond Pearl

Variable strategies

The two key variables in Verhulst’s equation are the maximum capacity of a species to reproduce (*r*), and the carrying capacity of the area (*K*). Organisms are either *r*-strategists or *K*-strategists. *R*-strategists, such as bacteria, mice, and small birds, reproduce rapidly, mature early, and have a